Category Theory for Mathematical Physics — Exercise Sheet 4

Duality: cups, caps and quantum dimension

Variant: Hints and answers

Conventions as in Lecture 4: $\mathrm{ev}_X\colon X^*\otimes X\to 1$ is a cap and $\mathrm{coev}_X\colon 1\to X\otimes X^*$ a cup (Definition 4.1); a pivotal structure a gives the traces of Definition 4.6; in $\mathrm{Vect}_{\mathrm{fd}}$, a is the canonical $V\to V^{**}$. In Exercise 2, for a qubit, $|0\rangle, |1\rangle$ is the standard basis, X and Z are the Pauli matrices (elsewhere X is an object), and $|\Phi^{\pm}\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$, $|\Psi^{\pm}\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$. The letter j is a gauge number in Exercise 4 and an isomorphism $V\to V^*$ in Exercises 5 and 6, as in Definition 4.9; spins are written ℓ, so the lecture's $(-1)^{2j}a$ is $(-1)^{2\ell}a$ here.

Exercise 1 (Rotated boxes, a second dual and the cyclic trace). (i) In $\mathrm{Vect}_{\mathrm{fd}}$ let $f\colon X\to Y$, with bases (x_k) of X and (y_l) of Y, dual bases (x^k), (y^l), and $f(x_k)=\sum_l f_{lk}\,y_l$. Show that the rotated box f^* of Equation (4.2) has the transpose matrix of f.

(ii) Let $X=\mathbb{C}^2$ and $G=\begin{pmatrix}1&1\\0&1\end{pmatrix}$. Take $X'=\mathbb{C}^2$ (column vectors) with $\mathrm{ev}'(u\otimes x)=u^{T}Gx$. Find coev′ making $(X',\mathrm{ev}',\mathrm{coev}')$ a left dual of X (check both snakes), and compute the isomorphism $\varphi\colon X^*\to X'$ of Theorem 4.2(a), with X^* written as row vectors.

(iii) Let $f=\begin{pmatrix}1&2&0\\0&1&3\end{pmatrix}\colon\mathbb{C}^3\to\mathbb{C}^2$ and $g=\begin{pmatrix}1&0\\0&1\\1&1\end{pmatrix}\colon\mathbb{C}^2\to\mathbb{C}^3$. Compute $\mathrm{tr}(g\circ f)$ and $\mathrm{tr}(f\circ g)$, and draw the three pictures of Equation (4.8) for this pair.

(iv) Cyclicity moves a box around the loop. Show that it does not allow other reorderings: find 2×2 matrices A,B,C with $\mathrm{tr}(ABC)\neq\mathrm{tr}(ACB)$. What, in the picture, stops C from jumping over B?

Hint. (i) Feed y^l into the picture: the cup creates $\sum_k x_k\otimes x^k$, f acts on x_k, and the cap pairs y^l with $f(x_k)$. (ii) Try $\mathrm{coev}'(1)=\sum_i e_i\otimes c_i$ and solve the first snake for the c_i. (iv) Try the matrix units E_{12}, E_{21}, E_{11}.

Answer. (i) $f^*(y^l)=\sum_k f_{lk}\,x^k$: the matrix f^{T}. (ii) $\mathrm{coev}'(1)=\sum_i e_i\otimes G^{-T}e_i=e_1\otimes(e_1-e_2)+e_2\otimes e_2$; $\varphi(\xi)=G^{-T}\xi^{T}$, with $G^{-T}=\begin{pmatrix}1&0\\-1&1\end{pmatrix}$. (iii) Both are 5. (iv) $A=E_{12}$, $B=E_{21}$, $C=E_{11}$: $\mathrm{tr}(ABC)=1$, $\mathrm{tr}(ACB)=0$; boxes on one strand cannot pass each other.

Exercise 2 (Bent wires and teleportation's four outcomes). (i) Draw the two maps of Equation (4.4), $\mathrm{Hom}(U\otimes V,W)\to\mathrm{Hom}(U,W\otimes V^*)$ and back, and show with one snake equation that bending a strand up and then down returns the morphism you started with. In $\mathrm{Vect}_{\mathrm{fd}}$ with $U=\mathbb{C}$, what does bending turn $f\colon\mathbb{C}^3\to\mathbb{C}^2$ of Exercise 1 into?

Alice holds $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$ and the first half of $|\Phi^+\rangle$; Bob holds the second half.

(ii) For each Bell state $|B\rangle\in\{|\Phi^+\rangle,|\Phi^-\rangle,|\Psi^+\rangle,|\Psi^-\rangle\}$ compute $(\langle B|\otimes\mathrm{id})(|\psi\rangle\otimes|\Phi^+\rangle)$, and write it as $\frac{1}{2}\sigma_B|\psi\rangle$.

(iii) Each Bell state is $(\mathrm{id}\otimes M_B)|\Phi^+\rangle$ for a 2×2 matrix M_B. Draw $\langle B|$ as the cap of Equation (4.5) with a box on its right leg, and explain σ_B by sliding that box along the snake to Bob's strand.

Hint. (i) Slide f along the strand it sits on; the snake left over is the one of Definition 4.1 for V. (ii) Expand $|\psi\rangle \otimes |\Phi^+\rangle$ in the Bell basis on Alice's two qubits. (iii) A box on one leg of the cup $|\Phi^+\rangle$ moves to the other leg as its transpose.

Answer. (i) Down after up gives f by the snake for V; f becomes the element $\sum_k f(x_k) \otimes x^k = \sum_{l,k} f_{lk}\, y_l \otimes x^k$ of $\mathbb{C}^2 \otimes (\mathbb{C}^3)^*$. (ii) $\sigma_B = I, Z, X, XZ$ for $\Phi^+, \Phi^-, \Psi^+, \Psi^-$. (iii) $M_B = I, Z, X, XZ$; $\langle B|$ carries $M_B^\dagger$, which reaches Bob as $(M_B^\dagger)^T = \overline{M_B} = M_B$ (the four are real).

Exercise 3 (Super vector spaces). The objects of sVec are the finite-dimensional $\mathbb{Z}_2$-graded spaces $V = V_0 \oplus V_1$; its morphisms are the linear maps preserving the grading. It has the tensor product with $(V \otimes W)_0 = V_0 \otimes W_0 \oplus V_1 \otimes W_1$, $(V \otimes W)_1 = V_0 \otimes W_1 \oplus V_1 \otimes W_0$, the unit $\mathbb{C}$ in degree 0, and the duals, ev and coev of $\mathrm{Vect}_{\mathrm{fd}}$, with $(V^*)_i = (V_i)^*$. Let P_V be $+1$ on V_0 and -1 on V_1.

(i) Show that $a'_V = a_V \circ P_V$ is a pivotal structure.

(ii) Show that its left trace is the *supertrace*, $\mathrm{str}(h) = \mathrm{tr}(h|_{V_0}) - \mathrm{tr}(h|_{V_1})$, so that $d_V = \dim V_0 - \dim V_1$, and that its right trace is also the supertrace.

(iii) Draw the picture proof of $d_{V\otimes W} = d_V d_W$ (Theorem 4.7): the loop on $V \otimes W$ as two nested loops. Then check $d_{V\otimes W} = d_V d_W$ directly from the grading of $V \otimes W$, and find a nonzero object of dimension 0.

Lecture 7 adds the braiding that makes this category "super"; Lecture 10 uses the pivotal structure a on it instead, with $d = +1$ on the odd line.

Hint. (i) A grading-preserving map commutes with P. (ii) Feed $\mathrm{coev}(1) = \sum_i e_i \otimes e^i$, with a graded basis, into the cap built from a'.

Answer. (i) a' is natural ($fP_V = P_W f$) and $P_{V\otimes W} = P_V \otimes P_W$. (ii) $\mathrm{tr}_L(h) = \sum_i e^i(P_V h e_i) = \mathrm{str}(h)$, and likewise on the right. (iii) The loop on $V \otimes W$ is the W loop nested inside the V loop; $d_{\mathbb{C}^{1|1}} = 0$ for $V_0 = V_1 = \mathbb{C}$.

Exercise 4 (The semion snake). In Vec_G^ω on $\mathbb{Z}_2 = \{1, s\}$ with $\omega(s,s,s) = -1$, $s^* = s$ and $s \otimes s = 1$, so a cup $\mathrm{coev}_s = \lambda\ \mathrm{id}_1$ and a cap $\mathrm{ev}_s = \mu\ \mathrm{id}_1$ are numbers.

(i) Show that the first snake of Equation (4.1) is $\lambda\mu\,\omega(s,s,s)\ \mathrm{id}_s$, so both snakes are straight exactly when $\lambda\mu = -1$. What is the snake for the naive choice $\lambda = \mu = 1$?

(ii) A *gauge* changes the identification $s \otimes s = 1$ by a number $j \neq 0$: it is the monoidal functor that is the identity on objects with $J_{s,s} = j$ and the other J's and J_0 equal to 1. It sends (λ, μ) to $(\lambda/j, \mu j)$. Show that it is a monoidal functor, and that no gauge makes $\lambda = \mu = 1$ a solution. Is there a solution with $\lambda = \mu$?

Hint. (i) The snake passes through one associator $\alpha_{s,s,s}$ (Example 4.4). (ii) Use the first condition of Definition 3.10 at (s,s,s), and watch the product $\lambda\mu$.

Answer. (i) The naive snake is $-\mathrm{id}_s$. (ii) The product $(\lambda/j)(\mu j) = \lambda\mu$ stays -1; $\lambda = \mu = \pm i$ is a solution.

Exercise 5 (Frobenius–Schur indicators of Q_8 and D_8). Let $Q_8 \subset \mathrm{SU}(2)$ be generated by $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and D_8, the symmetries of a square, by the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and the reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Each acts on $V = \mathbb{C}^2$ by these matrices, irreducibly.

(i) Show that both groups have order 8, and that the two characters on V take the value 2 once, -2 once and 0 six times.

(ii) Compute $\kappa_V = \frac{1}{8}\sum_g \chi_V(g^2)$ for each (the character formula of *Symmetries*).

(iii) Find a nonzero invariant bilinear form on V for each group, and check against Definition 4.9 that a nonzero symmetric invariant form gives $\kappa = +1$ and a nonzero antisymmetric one $\kappa = -1$. The two characters take the same values, with the same multiplicities; what group data, beyond these values, does κ see?

Hint. (ii) Count the elements whose square is the identity. (iii) D_8 consists of real orthogonal matrices; $Q_8 \subset \mathrm{SU}(2)$ fixes the form Ω of Theorem 4.8. A nonzero invariant form is nondegenerate, since its kernel is a subrepresentation.

Answer. (i) $Q_8 = \{\pm 1, \pm q_1, \pm q_2, \pm q_3\}$ with q_1, q_2 the generators and $q_3 = q_1 q_2$, and D_8 has 4 rotations and 4 reflections; $\chi = 2$ at 1, -2 at the central element, 0 elsewhere. (ii) $\kappa = -1$ for Q_8, $+1$ for D_8. (iii) Ω for Q_8, $v^T w$ for D_8; κ sees the squaring map $g \mapsto g^2$: 2 elements square to 1 in Q_8, 6 in D_8.

Exercise 6 ⋆. *The Penrose cap and the pivotal structure.* Let $V = \mathbb{C}^2$ be spin-$\frac{1}{2}$, with $\Omega(e_1, e_2) = 1 = -\Omega(e_2, e_1)$ as in Theorem 4.8, and $j\colon V \to V^*$, $j(v) = \Omega(\cdot, v)$.

(i) Show that $(\mathrm{id} \otimes j^{-1}) \circ \mathrm{coev}_V$ is the cup Ω and that $\mathrm{ev}_V \circ (j \otimes \mathrm{id})$ is the cap $-\Omega^\dagger$: the straight pair of Theorem 4.8 is (coev, ev) moved along j.

(ii) Show that the loop $-\Omega^\dagger \circ \Omega = -2$ is the quantum dimension $\mathrm{tr}_L(\mathrm{id}_V)$ for the map $a''_V\colon V \to V^{**}$ with $a''_V(v)(\varphi) = \Omega(j^{-1}\varphi, v)$, and that $a''_V = -a_V$.

(iii) On the finite-dimensional representations of $\mathrm{SU}(2)$ let $a'_W = a_W \circ \rho_W(-1)$. Show that a' is a pivotal structure, equal to a'' on spin-$\frac{1}{2}$; compute d_W for a' on spin ℓ, and the Frobenius–Schur indicator of spin-$\frac{1}{2}$ for a'.

Hint. (i) Compute $j(e_1)$ and $j(e_2)$ in the dual basis. (ii) The cap of (i), precomposed with $\mathrm{id} \otimes j^{-1}$, is a cap on $V \otimes V^*$; write it as $\mathrm{ev}_{V^*} \circ (a'' \otimes \mathrm{id})$, and evaluate $a''_V(v)$ on $\varphi = j(w)$. (iii) -1 is central in $\mathrm{SU}(2)$ and acts on spin ℓ by $(-1)^{2\ell}$.

Answer. (i) $j(e_1) = -e^2$, $j(e_2) = e^1$, so $(\mathrm{id} \otimes j^{-1})(\sum e_i \otimes e^i) = e_1 \otimes e_2 - e_2 \otimes e_1$. (ii) $a''_V(v)(j(w)) = \Omega(w, v) = -a_V(v)(j(w))$. (iii) $d_W = (-1)^{2\ell}(2\ell + 1)$, so -2 on spin-$\frac{1}{2}$; $\kappa = +1$.